CelebA - ten-seed IID and non-IID main tables

Main table | 10 shared seeds: 91001-91010 | mean +/- sample SD (ddof=1)

Best: bold. Second-best: underlined. Fairness ranking requires mean ACC >80%; dagger marks ineligible values. Displayed ties share rank.

Table 1. IID (alpha=5000)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	84.72 ± 11.74	0.0384 ± 0.0176	0.0937 ± 0.0366	89.57 ± 0.78	0.0519 ± 0.0096	0.1212 ± 0.0064	68.19 ± 5.77	0.1004 ± 0.0410†	0.0993 ± 0.0380†	66.01 ± 3.57	0.0950 ± 0.0477†	0.0807 ± 0.0413†	84.37 ± 2.38	0.0280 ± 0.0076	0.0693 ± 0.0161
FairFed	89.47 ± 0.85	0.0414 ± 0.0076	0.1102 ± 0.0077	89.79 ± 0.76	0.0489 ± 0.0042	0.1188 ± 0.0043	69.47 ± 4.65	0.1158 ± 0.0635†	0.1120 ± 0.0554†	62.59 ± 5.11	0.1594 ± 0.0822†	0.1461 ± 0.0668†	85.55 ± 2.31	0.0288 ± 0.0081	0.0781 ± 0.0116
Median	86.28 ± 1.31	0.0510 ± 0.0081	0.0980 ± 0.0117	86.52 ± 1.35	0.0610 ± 0.0087	0.1062 ± 0.0111	84.66 ± 1.47	0.0438 ± 0.0135	0.0825 ± 0.0142	85.00 ± 1.38	0.0484 ± 0.0090	0.0852 ± 0.0127	85.54 ± 1.50	0.0534 ± 0.0096	0.0965 ± 0.0124
FLTrust	89.43 ± 0.75	0.0432 ± 0.0075	0.1111 ± 0.0062	89.67 ± 0.82	0.0518 ± 0.0072	0.1207 ± 0.0064	**89.68 ± 0.74**	0.0422 ± 0.0069	0.1119 ± 0.0071	**89.68 ± 0.74**	0.0422 ± 0.0069	0.1119 ± 0.0071	**89.72 ± 0.64**	0.0496 ± 0.0061	0.1177 ± 0.0057
FedAA-DDPG	88.35 ± 1.33	0.0455 ± 0.0116	0.1083 ± 0.0100	89.06 ± 0.92	0.0442 ± 0.0137	0.1097 ± 0.0143	88.80 ± 1.06	0.0423 ± 0.0116	0.1071 ± 0.0115	88.82 ± 1.21	0.0385 ± 0.0131	0.1042 ± 0.0152	88.60 ± 1.12	0.0493 ± 0.0089	0.1136 ± 0.0095
LASA	89.39 ± 1.42	0.0377 ± 0.0049	0.1074 ± 0.0088	89.46 ± 0.79	0.0493 ± 0.0066	0.1176 ± 0.0059	85.75 ± 1.71	0.0330 ± 0.0109	0.0811 ± 0.0138	85.96 ± 1.62	0.0366 ± 0.0105	0.0839 ± 0.0152	88.20 ± 1.40	0.0450 ± 0.0073	0.1050 ± 0.0113
FairGuard	89.18 ± 0.83	0.0317 ± 0.0026	0.0998 ± 0.0062	89.57 ± 1.24	0.0369 ± 0.0071	0.1078 ± 0.0092	49.32 ± 1.59	0.0042 ± 0.0132†	0.0049 ± 0.0156†	50.33 ± 1.72	0.0000 ± 0.0000†	0.0000 ± 0.0000†	83.20 ± 1.76	0.0205 ± 0.0098	0.0491 ± 0.0199
FLTrust+FairGuard	**89.93 ± 1.10**	0.0365 ± 0.0075	0.1074 ± 0.0113	**90.37 ± 0.60**	0.0363 ± 0.0086	0.1118 ± 0.0100	77.83 ± 12.15	0.0298 ± 0.0299†	0.0607 ± 0.0354†	75.99 ± 16.52	0.0293 ± 0.0146†	0.0652 ± 0.0419†	86.44 ± 5.76	0.0368 ± 0.0144	0.0903 ± 0.0325
LoGoFair-DP	83.64 ± 11.34	0.0453 ± 0.0189	**0.0071 ± 0.0059**	88.03 ± 0.79	0.0564 ± 0.0077	**0.0079 ± 0.0040**	69.18 ± 5.69	0.0161 ± 0.0115†	0.0103 ± 0.0071†	66.76 ± 3.39	0.0129 ± 0.0106†	0.0100 ± 0.0055†	83.61 ± 2.26	0.0284 ± 0.0115	**0.0077 ± 0.0061**
FLGMM	89.33 ± 0.99	0.0404 ± 0.0065	0.1074 ± 0.0073	89.72 ± 0.83	0.0458 ± 0.0078	0.1171 ± 0.0049	89.34 ± 1.01	0.0424 ± 0.0070	0.1113 ± 0.0081	89.34 ± 1.01	0.0423 ± 0.0075	0.1113 ± 0.0080	89.49 ± 0.93	0.0460 ± 0.0061	0.1143 ± 0.0063
GuardFed-AD2+	88.26 ± 1.04	**0.0097 ± 0.0075**	0.0625 ± 0.0138	88.39 ± 0.63	**0.0107 ± 0.0096**	0.0607 ± 0.0103	88.47 ± 0.91	**0.0061 ± 0.0042**	**0.0640 ± 0.0142**	88.69 ± 0.81	**0.0077 ± 0.0046**	**0.0638 ± 0.0070**	88.38 ± 0.89	**0.0165 ± 0.0072**	0.0559 ± 0.0156

Table 2. non-IID (alpha=5)

Method	Benign			F Flip			FedSA			S-DFA			Sp-DFA		
	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD	ACC (%)	AEOD	ASPD
FedAvg	86.85 ± 5.45	0.0428 ± 0.0137	0.0958 ± 0.0353	89.51 ± 1.82	0.0492 ± 0.0045	0.1161 ± 0.0143	74.76 ± 13.38	0.0644 ± 0.1097†	0.0614 ± 0.0893†	74.51 ± 13.37	0.0522 ± 0.0913†	0.0591 ± 0.0756†	85.48 ± 2.58	0.0267 ± 0.0157	0.0741 ± 0.0247
FairFed	89.59 ± 0.97	0.0388 ± 0.0057	0.1092 ± 0.0071	89.83 ± 0.92	0.0462 ± 0.0081	0.1163 ± 0.0071	71.45 ± 11.85	0.0759 ± 0.0860†	0.0730 ± 0.0885†	67.06 ± 11.91	0.1307 ± 0.1184†	0.1171 ± 0.1067†	86.18 ± 2.41	0.0328 ± 0.0133	0.0829 ± 0.0204
Median	86.22 ± 1.31	0.0518 ± 0.0068	0.0963 ± 0.0096	86.51 ± 1.24	0.0594 ± 0.0084	0.1067 ± 0.0097	84.47 ± 1.30	0.0425 ± 0.0118	0.0799 ± 0.0113	84.48 ± 1.35	0.0461 ± 0.0142	0.0839 ± 0.0124	85.47 ± 1.26	0.0502 ± 0.0111	0.0929 ± 0.0086
FLTrust	89.68 ± 0.60	0.0445 ± 0.0049	0.1132 ± 0.0060	89.83 ± 0.69	0.0493 ± 0.0054	0.1197 ± 0.0044	**89.48 ± 0.70**	0.0441 ± 0.0063	0.1120 ± 0.0041	**89.48 ± 0.70**	0.0441 ± 0.0063	0.1120 ± 0.0041	**89.64 ± 0.71**	0.0475 ± 0.0052	0.1153 ± 0.0053
FedAA-DDPG	88.60 ± 1.14	0.0456 ± 0.0125	0.1069 ± 0.0151	88.56 ± 1.19	0.0491 ± 0.0132	0.1110 ± 0.0103	88.53 ± 1.31	0.0421 ± 0.0231	0.1031 ± 0.0212	88.52 ± 1.60	0.0409 ± 0.0122	0.1020 ± 0.0149	88.39 ± 1.14	0.0597 ± 0.0189	0.1188 ± 0.0190
LASA	89.27 ± 0.85	0.0386 ± 0.0057	0.1070 ± 0.0084	89.51 ± 0.88	0.0500 ± 0.0096	0.1181 ± 0.0059	85.91 ± 1.69	0.0301 ± 0.0069	0.0767 ± 0.0129	85.93 ± 1.64	0.0345 ± 0.0113	0.0791 ± 0.0128	88.15 ± 1.22	0.0417 ± 0.0037	0.1026 ± 0.0074
FairGuard	89.39 ± 1.11	0.0314 ± 0.0054	0.1026 ± 0.0086	89.84 ± 0.76	0.0394 ± 0.0103	0.1109 ± 0.0090	59.95 ± 16.51	0.0066 ± 0.0094†	0.0169 ± 0.0251†	61.29 ± 15.59	0.0108 ± 0.0161†	0.0206 ± 0.0317†	78.37 ± 14.94	0.0209 ± 0.0104†	0.0552 ± 0.0238†
FLTrust+FairGuard	**89.69 ± 1.68**	0.0357 ± 0.0055	0.1085 ± 0.0102	**90.07 ± 1.34**	0.0426 ± 0.0062	0.1144 ± 0.0088	81.90 ± 11.66	0.0400 ± 0.0214	0.0851 ± 0.0350	80.44 ± 12.06	0.0359 ± 0.0262	0.0763 ± 0.0174	79.69 ± 13.94	0.0408 ± 0.0124†	0.0892 ± 0.0276†
LoGoFair-DP	85.64 ± 4.92	0.0445 ± 0.0232	**0.0055 ± 0.0051**	87.96 ± 1.66	0.0535 ± 0.0147	**0.0060 ± 0.0036**	74.94 ± 12.90	0.0192 ± 0.0147†	0.0084 ± 0.0054†	74.53 ± 12.82	0.0196 ± 0.0178†	0.0090 ± 0.0061†	84.53 ± 2.27	0.0358 ± 0.0122	**0.0051 ± 0.0045**
FLGMM	89.02 ± 0.91	0.0396 ± 0.0066	0.1071 ± 0.0078	89.28 ± 0.98	0.0525 ± 0.0077	0.1185 ± 0.0068	89.15 ± 0.88	0.0428 ± 0.0049	0.1088 ± 0.0055	89.17 ± 1.15	0.0423 ± 0.0048	0.1083 ± 0.0105	**89.64 ± 0.72**	0.0446 ± 0.0084	0.1147 ± 0.0053
GuardFed-AD2+	88.59 ± 1.17	**0.0070 ± 0.0042**	0.0651 ± 0.0090	88.44 ± 0.74	**0.0100 ± 0.0070**	0.0589 ± 0.0092	88.41 ± 0.92	**0.0078 ± 0.0056**	**0.0616 ± 0.0141**	88.22 ± 0.78	**0.0126 ± 0.0093**	**0.0584 ± 0.0189**	88.34 ± 0.96	**0.0087 ± 0.0066**	0.0596 ± 0.0146

Reading: Higher ACC and lower AEOD/ASPD are better. Best/second-best eligible displayed means are **bold**/underlined; displayed ties share rank. † retains a disparity whose corresponding mean ACC is <=80%, excluding it only from fairness ranking. ACC ranking and all seeds/means/sample SDs are unchanged. Eligibility uses unrounded mean ACC; SD does not rank. This post-hoc display rule was author-approved on 2026-10-12, not preregistered. Marks do not establish significance or undefined metrics. ACC is percent; gaps are [0,1]; AEOD is the absolute TPR gap, not full equalized odds. Shading identifies GuardFed.

Method scope: Native predictions preserve each method's own postprocessing; this does not isolate aggregation effects. FairFed/FairGuard/composition, FedAA and LASA are documented adaptations. LoGoFair-DP uses fitted demographic-parity postprocessing, 20 image-ID virtual cohorts (not training clients), fit seed 1719 and 30 postprocessing rounds. FLGMM uses the author-code full-model/largest-cluster adaptation and native uncalibrated outputs.

Coverage and provenance: Eleven methods, two distributions, five conditions and 10 seeds give 1100 evaluation records, not 1100 independent training models: LoGoFair reuses FedAvg checkpoints. Backbone checkpoints are terminal round 70. FLGMM contributes 96 new and 4 explicitly reused screen records; its selected recipe is warmup 20/control width 2.0/local learning rate 0.001. Original mixed training/replay environments and separate CPU saved-state verification remain disclosed; no cross-runtime equivalence is claimed.

Selection boundary: Seed 91001 participated in validation selection; other validation seeds and historical test metadata were exposed. The archived fixed 9/6-seed sensitivity views remain available separately. No score ranking, significance, final-test or complete 17-method claim is made. This display does not select the manuscript's final primary endpoint.

Source: root-adopted eleven-method native1100 display table; no new statistics, fits, inference or training.